

Azure Virtual Machine Deployment Options (Azure Portal)

1. Basics

Option	Description
Subscription	Select billing account.
Resource Group	Logical container for resources.
VM Name	Unique identifier for the VM.
Region	Choose datacenter location.
Availability Options	- No redundancy - Availability Zone - Availability Set
Image	OS selection (Windows, Linux, custom).
VM Size	CPU, RAM, and disk configuration.

2. Administrator Account

Option	Description
Username	Admin login name.
Authentication Type	Password or SSH Key (recommended for Linux).

3. Disks

Disk Type	Performance	Cost	Use Case
Standard HDD	Low	Low	Dev/test workloads
Standard SSD	Moderate	Medium	Balanced performance/cost
Premium SSD	High	High	Production, mission-critical apps

- **Encryption:** Platform-managed or customer-managed keys
 - **Data Disks:** Attach additional storage
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4. Networking

Diagram:

```
[Virtual Network]
|
[Subnet]
```

Author: Prabu Thiagamani

|
[NIC + NSG]
|
[Public IP] → Internet

- **VNet:** Logical network boundary
 - **Subnet:** Segment within VNet
 - **NSG:** Firewall rules
 - **Public IP:** Optional external access
 - **Load Balancer:** Distribute traffic across VMs
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5. Management

- **Monitoring:** Boot diagnostics, guest diagnostics
 - **Identity:** System-assigned or user-assigned managed identity
 - **Auto-shutdown:** Schedule VM shutdown
 - **Backup:** Enable Azure Backup
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6. Advanced

- **Extensions:** Install agents (monitoring, security)
 - **Custom Data / Cloud-init:** Initialization scripts
 - **Proximity Placement Group:** Optimize latency between VMs
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7. Tags

- Key-value pairs for organization
 - **Example:** Environment=Production, Owner=ITTeam
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8. Review + Create

- Final validation of all settings
 - Deployment begins once confirmed
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